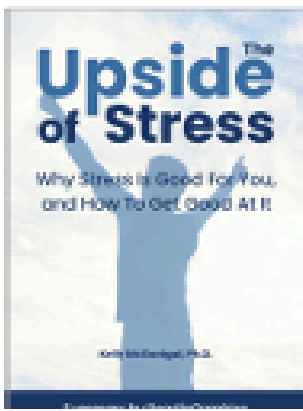




THE UPSIDE OF STRESS

Why Stress Is Good for You,
and How to Get Good at It

KELLY MCGONIGAL, PH.D.



The Big “So What”

Most people believe that stress is bad and should be avoided. In this book, health psychologist Kelly McGonigal presents scientific evidence that stress can improve your performance, physical and psychological health. She also shares various strategies and interventions to help you thrive under pressure, and manage physical symptoms of anxiety.



Introduction



Like many others, McGonigal used to think that stress is toxic. However, after researching stress in detail—through empirical research, interviews with scientists and researchers, experiments and interventions—she realized stress is harmful only when you believe it is.

- For example, 30,000 adults in the U.S. were asked if they thought stress hurt their health. 80 years later, those with high stress levels were 43% more likely have died—*only if they had believed that stress was bad for them.* Those who experienced high stress but didn't see it as harmful actually had the *lowest risks of death.* In short, stress alone doesn't kill people; the danger comes from the combination of stress and the belief that it's harmful.
- McGonigal has since combined psychology, neuroscience, and biology to create the “New Science of Stress” course being taught at Stanford, to help people thrive under stress. This book is based on the insights and strategies from the course, presented in 2 parts:
 - (i) How to change the way you think about stress; and
 - (ii) On-the-spot strategies to apply when you're stressed.

KEY QUOTES

“The latest science reveals that stress can make you smarter, stronger, and more successful. It helps you learn and grow. It can even inspire courage and compassion.”

“I’ve changed my mind about stress, and now I want to change yours.”



Part 1: Rethinking Stress

The first step to transforming stress is to change how you think about it. The word “stress” is used by people to describe everything from a minor frustration to a major trauma. McGonigal defines stress as “what arises when something you care about is at stake.”



The Power of Mindsets



Many research studies have found that our beliefs can affect our physical outcomes:

- In a study, housekeepers who were told that their work was a form of exercise lost weight and body fat after 4 weeks. Housekeepers who were only told the importance of physical exercise didn't experience such weight/fat loss.
- In another study, 2 groups of people were given the same milkshake. However, the 1st group thought it was a high-calorie, high-fat milkshake, while the 2nd group thought it was a low-calorie, zero-fat milkshake. After consuming the milkshake, the 1st group was found to have 3x less hunger hormones (ghrelin), i.e. they felt full because they *expected* to be full.
- Beliefs about aging were found to affect people's rate of recovery after a health challenge or accident, and even their longevity. People with a positive attitude toward aging tend to live longer.



Mindsets are your core beliefs about how the world works, e.g. “the world is a dangerous place”, “money can't buy you happiness”, or “people can't change”.

- Your mindset affects how you think, feel and act, which influence your outcomes, to further reinforce your mindset. So, even a short, one-time mindset intervention can catalyze changes to produce a snowball effect with far-reaching consequences.

KEY QUOTES

“Small shifts in mindset can trigger a cascade of changes so profound that they test the limits of what seems possible.”

- For example, a group of freshmen received a mindset intervention targeting their sense of social belonging. They were told that it's normal to feel like you don't belong, and the feeling will fade. Then, they were asked to read excerpts from others' experiences, write an essay on their own experience, and read their essay in front of a camera. In the next 3 years, students who received this intervention experienced better physical health, academic performance and happiness, compared to those who didn't. Most of them forgot the details of the intervention, but the message stuck and influenced their behaviors (e.g. how they handled problems or connected with others) to create a long-term, ripple effect.
- Mindset interventions are similar to placebos, except the effect is longer-lasting. Placebos are effective even when patients know that they're taking sugar pills instead of real drugs, if the doctors explain how placebos work. Likewise, mindset interventions work even when participants know they're going through an intervention.



Your thoughts/beliefs about stress affect your physical and biological realities.

- During times of stress, our body releases 2 hormones—cortisol and dehydroepiandrosterone (DHEA).
 - (i) When you're stressed, your body burns fat and sugar for energy. *Cortisol* enhances your ability to use that energy, but also suppresses certain biological functions (e.g. digestion, reproduction, growth). *DHEA* helps your brain to learn from the stressful experience, and also counters some of cortisol's effects, e.g. it strengthens your immune system and speeds up wounds recovery.
 - (ii) You need both hormones, but the ratio of DHEA to Cortisol (called the *growth index*) affects the long-term impact of stress. People with a higher growth index tend to thrive under stress, and are less prone to stress-related diseases like depression or heart disease. Those with a lower growth index are more prone to depression, poorer immunity, etc.
- In a study, participants went through an interview designed to generate stress. Before the interview, each

participant watched a video that either said (i) stress enhances performance or (ii) stress is debilitating. After the stressful interview, both groups saw an increase in cortisol levels. However, the participants from the 1st group had higher DHEA levels compared to the 2nd group. In short, the belief that stress is enhancing improved their *growth index*.



The Biology and Psychology of Stress



The flawed belief that “stress is bad” can be traced back to Hans Selye, an endocrinologist. In 1936, Selye found that when lab rats were exposed to trauma (e.g. injected with substances, exposed to extreme heat/cold), they developed ulcers, physically deteriorated and died. He called this “stress”.

- Later, Selye hypothesized that stress also contributed to many symptoms in human patients (e.g. loss of appetite, fatigue) and ailments (e.g. heart attacks, allergies). He defined stress as “the response of the body to any demand made on it”—a definition that much broader than the extreme trauma/abuse experienced by the lab rats. In short, Selye’s hypothesis was never proven, and his conclusions were drawn from lab animals under drastically different conditions from our daily human stresses.
- Selye’s work on the toxicity of stress was so widely acclaimed that he was nominated 10x for the Nobel Prize. He later acknowledged that that not all stress is bad, and even started to differentiate between good stress (eustress) and bad stress (distress). However, the label “stress is bad” had stuck.



It’s also wrongly assumed that stress will always trigger the “fight-or-flight response”. This term was first coined by physiologist Walter Cannon in 1915, to describe the state of heightened sympathetic activation when animals faced a live-threatening situation. This response may also be triggered in humans. However, it’s just one of humans’ many possible stress responses. Your body responds to stress in complex ways, depending on (i) the nature of the

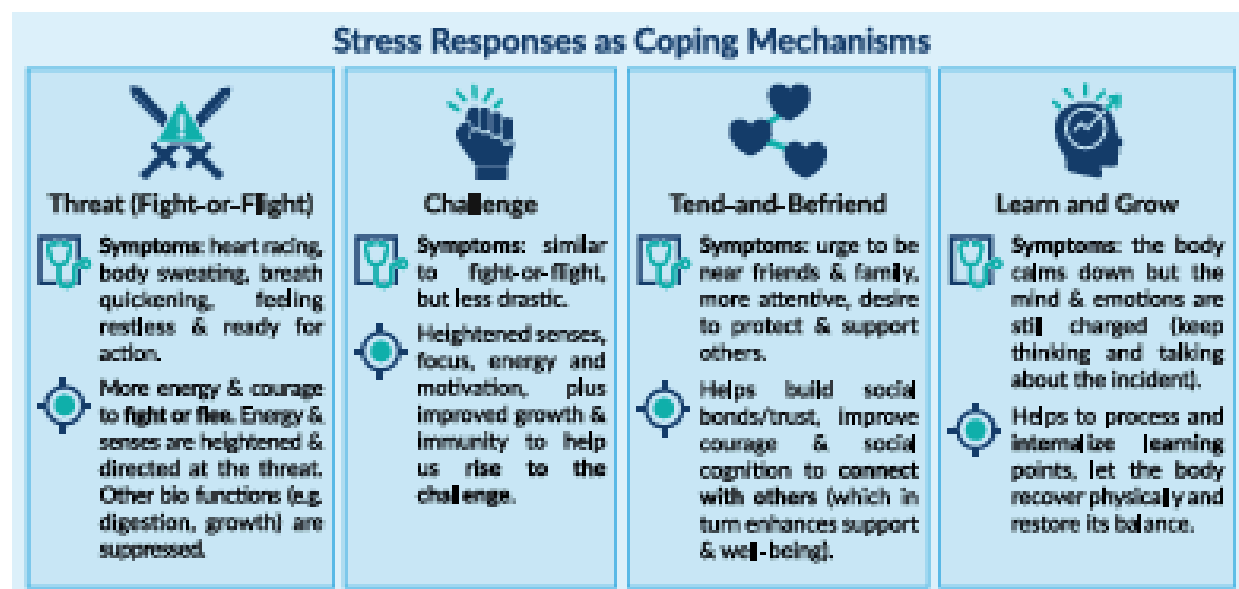
stress, (ii) your past experiences, and (iii) your interpretation of the situation.

- When you're stressed, the body unlocks energy and heightens your senses/focus to help you cope with the situation.
 - (i) When a *fight-or-flight* response kicks in, your breathing deepens to give you more oxygen, your body burns more fat and sugar for energy, the stress hormones facilitate the use of that energy, and your heart rate increases to deliver the oxygen/energy to your muscles/brain. Your pupils dilate and your hearing sharpens to heighten your focus. You also get a boost of motivation, power and confidence from biochemicals like endorphins, adrenaline, testosterone, and dopamine. In this state, you're driven by primal instincts instead of rational thought. The response can be so drastic that people have been known to exhibit extraordinary strength, e.g. 2 teenage girls managed to lift a 3000-pound tractor off their father.
 - (ii) However, in less stressful situations, your body usually adopts a moderate *challenge response*. You'll still have increased energy and focus, but your DHEA levels will be higher than your cortisol levels. Instead of operating from fear, you enter a growth state. People in a state of "flow" (i.e. enjoyably absorbed in a challenging task) are actually displaying a challenge response.
- Stress may also trigger your desire for social connection or the *tend-and-befriend response*. This comes mainly from oxytocin, which is also known as the "love hormone". Oxytocin helps to build social bonds, empathy, and trust. It also counters the debilitating effects of fear, improves cardiovascular health, and gives you the motivation/courage to protect others. So besides the desire to connect with people, you're more likely to stand up for a colleague or protect your loved ones. After a traumatic event or personal crisis, people tend to become more caring and altruistic because caring for others creates a sense of courage and optimism, and dispels feelings of fear and defeat.

KEY QUOTES

"Your stress response has a built-in mechanism for resilience—one that motivates you to care for others while also strengthening your physical heart."

- Stress also helps you to learn and grow. As your body starts to calm down, your stress hormones help you to recover faster. For example, cortisol and oxytocin help to reduce inflammation and restore balance to the nervous system, while DHEA helps to increase the brain's neuroplasticity to help you learn from the experience. That's why we tend to feel mentally and emotionally charged for some time after a stressful event. You're likely to replay the scene in your head, or feel the urge to tell someone about the incident.



 In short, stress isn't "bad" and stress hormones are your body's way of helping you to cope with stressful situations. Survivors of an accident are less likely to develop post-traumatic stress disorder (PTSD) if they have a higher level of stress hormones (cortisol and adrenaline) just after the accidents. Patients who received stress hormones just before a traumatic surgery spent less time in intensive care and had fewer traumatic symptoms post-surgery.

 In fact, stress goes hand-in-hand with a meaningful life.

- A Gallup World Poll found that a country's stress index was correlated with its people's well-being (life expectancy, GDP, happiness and life satisfaction). In fact, high levels of stress were associated with *both* higher levels of distress (including anger, sadness, worry etc.) and well-being (e.g. joy, love, laughter). McGonigal calls this the stress paradox. For example, parents with children below 18 are more stressed, but also laugh and smile more. Entrepreneurs are both stressed and fulfilled by their steep learning curve.

KEY QUOTES

"Stress leaves an imprint on your brain that prepares you to deal with similar stress the next time you encounter."

"The stress response is more than a basic survival instinct. It is built into how humans operate, how we relate to one another, and how we navigate our place in the world."

"A meaningful life is a stressful life."

- So, stress isn't a sign that something's wrong with your life; it's a sign that you're meaningfully engaged in things you care about. People who're busier tend to be happier, and people with a sense of purpose tend to live longer.
- In a study, students were tasked to keep journals over a winter break. One group wrote about their most important values (and how their daily activities supported those values). The other group wrote about the good things that happened to them. At the end of the break, the first group reported better health, confidence and motivation. Writing about their values helped them to see their daily activities as a expression of their values instead of burdens/hassles. Similar results were found in many other studies: by changing how you see your daily activities, you can motivate positive action and improve your self-identity.



Using Interventions to Change your Stress Mindset



Your stress mindset (whether you believe stress is harmful or helpful) tends to create a self-fulfilling prophecy. When you think of stress as a resource and actively confront your challenges, you develop new skills and resources to improve your results, competence and confidence in a virtuous cycle. When you deny or avoid the stressful challenges, the reverse happens.

- Psychologist Alia Crum did a stress mindset intervention with financial workers during the 2008 financial crisis. Participants were split into 3 groups. The 1st group was taught the positive effects of stress, the 2nd group was taught the harmful effects of stress, and the 3rd group received no training. Participants were surveyed before and after the training. After the intervention, people in the 1st group had a more balanced view about stress, were less depressed/anxious, reported fewer health issues and enjoyed more focus, engagement, collaboration, and productivity. The other 2 groups showed no behavioral changes.
- In another study, participants were invited to a stress-management training. Half the participants received a intervention while the rest were put on

KEY QUOTES

"Happy lives are not stress-free, nor does a stress-free life guarantee happiness."

"Stress and meaning are inextricably linked. You don't stress out about things you don't care about, and you can't create a meaningful life without experiencing some stress."

waitlist. During the intervention, participants were told the harmful and beneficial effects of stress, the power of mindset, and that the training aimed to help them choose a positive mindset. They were then asked to reflect on their own experiences with stress, and taught a 3-step process to apply when they were stressed. 3 weeks later, the participants demonstrated a more balanced view about stress, reported less anxiety/depression, and had better health, focus and engagement. The effects remained after 6 weeks. Employees on the waitlist showed no changes until they went through the intervention.



You can shape your stress response and transform your mindset using similar interventions or exercises:

- Rethink your stress response: Think of a recent stressful experience. Refer to the list of stress responses above, and write down the elements that were present during/after the experience, e.g. sweaty palms, the urge to call a friend. What did you associate with these elements? What helpful role could they have played?
- Reflect on the link between stress and meaning in life: List down the most meaningful aspects of your life (e.g. relationships, activities, goals). What feelings do they bring? Are they also somewhat stressful? Would you rather be stress-free or lose that aspect of your life?
- Consider your values. List down the values that are the most important to you, identify the top 3, then write about at least one of them for 10min. Describe why the value is important, how it guides your decisions, and how you express it in your daily life. The next time you're stressed, use this to remember why you do what you do. You can also carry a physical reminder, e.g. a keychain or post-it note with your key values.



Instead of complaining about stress or fantasizing about a stress-free life, try talking about stress to cultivate mindset mindfulness.

- When we avoid stress or try to forget the stressful moments, we're also cutting out the meaningful life experiences, creating additional stress while removing the resources that'd have helped us to cope. Physicians who shared their work experiences daily became more mindful

KEY QUOTES

"Remembering your values can help transform stress from something that is happening against your will and outside your control to something that invites you to honor and deepen your priorities."

"Although avoiding stress can seem like a rational strategy, it almost always backfires."

of their thoughts/feelings and found new meaning in their struggles at work. After 1 year, the physicians' depression and anxiety levels fell by two-thirds, and they felt more empathy for their patients.

- Reflect on the cost of stress avoidance: What roles/activities have you rejected because they seemed stressful? What substances/distractions do you use to escape unpleasant thoughts/feelings? What would you do/experience/change if you weren't afraid? What does avoiding stress truly cost you?



Part 2: Transforming Stress



Stress is unavoidable, and the strategies in this section will help you to transform stress from a debilitating factor to a useful resource. Being good at stress isn't about avoiding it or being untouched by it. It's about using stress to tap into your natural strengths and resources, so you're more likely to adopt a constructive response, instead of becoming withdrawn, aggressive or defensive.

- During times of stress, pause to consider what you need most, e.g. to survive, perform, connect, or learn. Your body will respond based on your expectations.



Transforming Threats to Challenges



A challenge response is better than a flight-or-flight response (or threat response) for performance enhancement. Not only does it increase your energy and focus, it also improves your performance, resilience and health.

- If you think that you lack the resources to deal with a challenge, you're more likely to feel threatened and lapse into a threat response. To shift to a challenge response, focus on your resources, including your strengths, past successes/preparation, and your body's natural stress responses.

KEY QUOTES

"When you stop resisting it, stress can fuel you."

*"If you have butterflies in your stomach, invite them into your heart."
—Cooper Edens*

"Whatever the sensations of stress are, worry less about trying to make them go away, and focus more on what you are going to do with the energy, strength, and drive that stress gives you."

- The Trier Social Stress Test used events like public speaking and math tests to create stressful situations that increased participants' stress hormones by up to 400%. A mindset intervention was applied before the test: the 1st group was told that stress could be used to boost performance, the 2nd group was told that stress was best ignored, and the last group received no tips about stress. During the tests, the 1st group was more confident, demonstrated more physiological stress arousal, better focus and performance. The last 2 groups showed no difference in performance.



You can transform anxiety into a resource for performance.

- Students who were told that pre-exam anxiety can help them to perform ended up doing better in a mock exam despite higher levels of stress activation, i.e. they were able to leverage their stress.
- The next time you notice signs of stress (e.g. racing heart, sweaty palms), don't force yourself to relax or suppress your anxiety. Instead, direct the energy constructively:
 - (i) Remind yourself that stress is your body's way of equipping you to cope with the challenge.
 - (ii) Tell yourself that you're excited or motivated, and use that energy to rise to the challenge.

This approach works for various situations and people, including the under-privileged and those with psychological disorders.



Building Resilience through Social Connections



Stress can bring about caring-befriending behaviors or aggressive-selfish behaviors, though the former is much more helpful.

- Tending and befriending helps to transform stress by increasing 3 types of brain activity to make us more social, brave and smart.
 - (i) Oxytocin makes us more empathic, trusting and sociable;

- (ii) Dopamine gives us a "rush" that dampens fear, increases optimism, and primes us for action; and
- (iii) Serotonin increases our sense of perception, intuition and self-control to help us identify the right solutions.



To switch to a tend-and-befriend mindset:

- Focus on bigger-than-self goals that positively impact others, and consider how your current activities can serve those goals.
 - (i) People who focus on a higher purpose tend to feel more inspired, curious, grateful and energized, build stronger networks and feel more satisfied with life. Those who focus on self-centered goals (e.g. promotions or awards) tend to feel more confused, anxious, envious and lonely.
 - (ii) A group of bus drivers reimagined their role to make the bus a place for people to feel seen and known. When passengers boarded the bus, the drivers greeted them, made eye contact and said hello. This simple shift significantly increased the drivers' job satisfaction and reduced risks of burnout.
 - (iii) Students who took time to reflect on bigger-than-self goals showed improved academic motivation and performance, and persisted longer in face of challenges.
- Look for ways to help/support others. By giving your time/help instead of withholding it, you'll actually feel less overwhelmed.
 - (i) By measuring human brain activity, researchers found that people who comfort/help others also experience hope and connection, whereas those who focus on removing their fears only became more fearful.
 - (ii) Time-strapped people who spent their free time helping others ended up feeling less time-constrained, more capable and useful, compared to others who used the free time on themselves. After a spouse's death, people are less prone to depression if they take care of others. Survivors of a natural disaster are less

KEY QUOTES

"Helping others doesn't make you live forever, and it can't protect you from everything. But what it does protect you from is stress."

prone to PTSD if they help others in the aftermath. Even prison inmates thrived when they were given caregiving opportunities.

- If you isolate yourself when you're stressed, you can get stuck in a downward spiral. To make a mindset shift, recognize your pain as part of a common human experience:
 - (i) In a group, get everyone to write down something they've been secretly struggling with. Then, mix the entries up, and get each person to read a random entry aloud. This allows us to see others' invisible pain, while protecting their anonymity.
 - (ii) On your own, reflect on your thoughts/feelings (e.g. pain, anger, self-doubt) about a situation. Consider how others may also feel similar pain/suffering. End the reflection with a phrase that captures your shared experience, e.g. "May we all know our own strength" or "May we all find peace."
- Take the first step to create the supportive community you wished you had. Don't wait for others to reach out first. If you wish to feel appreciated, then start by expressing your gratitude. If you want a sense of belonging, then help others to feel welcome.



Growing from Adversity



There's extensive evidence of mankind's ability to grow through adversity. The most significant personal growths in your life probably also occurred during times of stress.

- There are many stories of people who experienced extreme trauma yet rose above it. Post-traumatic growth has been observed in all sorts of trauma, e.g. abuse, terrorist attacks, or life-threatening diseases. Survivors may transform their lives as they uncovered inner strengths, experienced spiritual growth, found a new calling, or found deep, meaningful relationships.
- Trauma is never good, but it can activate one's strengths, values, and responses to bring good outcomes. We

KEY QUOTES

"Sometimes, we make the choice to be generous first, and the uplift comes later."

can't avoid pain and adversity in life, but we can choose our responses to them.



Psychologist Edith Chen found that people with a difficult childhood could resist the toxic environment with a "shift-and-persist" strategy. *Shifting* is about changing how you think about the stressful situation, e.g. believing that good can come out of it, or focusing on what you can learn. *Persisting* is about finding meaning in adversity, e.g. believing that things will get better or you have a purpose on earth. To adopt this strategy:

- Think of a stressful experience where you persevered or grew. Take 15min to write about the strengths/resources you drew on, the actions you took, how you dealt with adversity, and how you grew stronger.
- Think of a challenge you're struggling with. Consider the strengths/resources you can draw on, the skills/strengths you wish to develop, and how you can use this situation as an opportunity to learn and grow.



You can choose to see the upside in any situation. Yet, it's hard to do so when you're in the midst of a stressful situation, e.g. if you've been diagnosed with cancer. Train yourself to pause and find benefits in a stressful situation, so you can shift your physical, mental and emotional states and activate effective coping strategies.

- Take 2 minutes to consider the potential benefits of your current situation, e.g. how you could learn, grow or become stronger from it. Common benefits from adversity include: appreciation of life, spiritual growth, sense of personal strength, social connections, or new life directions. This 2-min mindset intervention can activate parts of the brain associated with positive motivation and active-coping, to help people to feel less angry, more joyful, grateful, and in control.
- Longer-term interventions include: daily reflections or journaling about your difficult situations, or keeping a diary of your positive daily experiences.



Spreading growth and resilience. Research found that watching the news after a natural disaster or terrorist attack tends to increase the risk of depression or PTSD. Even

KEY QUOTES

"We cannot always control the stress in our lives, but we can choose our relationship to it."

watching TV talk shows can increase fear and anxiety levels. Yet, research also found that people can be inspired by stories of others who experienced amazing resilience and recovery in face of trauma and adversity.

- We can all do our part to spread stories of resilience and growth, e.g. via news reports, artwork, pictures, newsletters, websites and conversations.
- Try capturing your own story. If you were a journalist, how would you describe your challenges, the turning point in your struggles, how you expressed your strength, resilience, growth and values, and how you changed/grew from adversity? What photos, videos or collages would best capture that story? You can keep the materials for private reference or share them.



Conclusion and Other Details in the Book

Stress can be harmful or helpful, depending on how you choose to see it. It's the most harmful if you (i) feel inadequate to handle it, (ii) isolate yourself from others and (iii) feel helpless or burdened by it. However, when you embrace it as a natural human resource, it can be a source of confidence, courage, connection, and meaning. The book comes with a wide range of empirical research, studies and interventions to elaborate on the concepts outlined in this summary. For more details, please visit <http://kellymcgonigal.com/books>.

About the Author



Kelly McGonigal, Ph.D., is a health psychologist and lecturer at Stanford University. She's best known for her work in the area of "science help", which combines insights from psychology, neuroscience, medicine, and biology to provide strategies that support health and well-being. She's the former editor in chief of the International Journal of Yoga Therapy, as well as the author of several books on health and wellness.

KEY QUOTES

"The best way to manage stress isn't to reduce or avoid it, but rather to rethink and even embrace it."